Generative AI and the Online Information Ecosystem: Current Evidence and Implications

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Workshop on AI and Political Economy of Information Filtering
Oxford University, 25th-27th May 2026

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Overview of My Research

Media Platforms and Technology

(1) Rise of Online Platforms:
Ad revenue, media bias and Craigslist

(2) Democratization of Content Creation and Distribution: Social media and political donations

(3) (Gen)AI and the information ecosystem
-- Current focus

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Overview

- *Tutorial lectures*: “The goal of these lectures is to bring all participants on the same page, especially with regards to the relevant state of research in other sub-fields.”

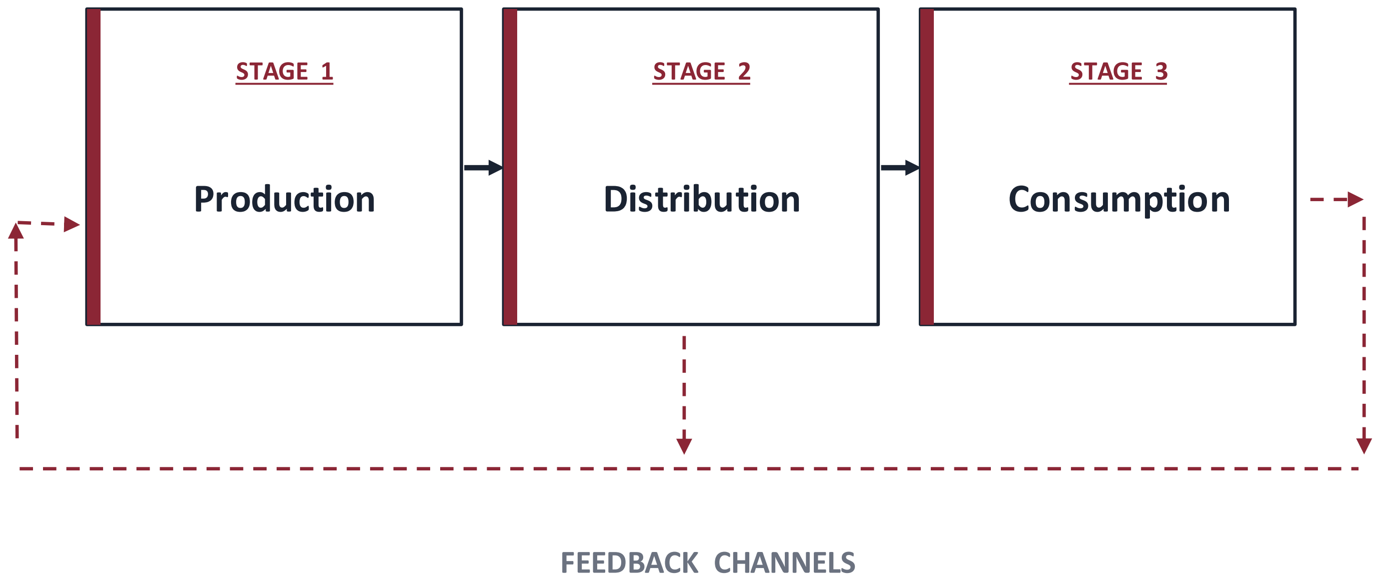
- Used this as an opportunity to learn about the literature in a (semi) structured way.
- The list of papers are not exhaustive by any means (and its not meant to be).
- The entire hope is to get some conversation started.
- I’ll inject my own takes from time to time.

“Defining” Terms: GenAI and Online Information Ecosystem

- **Generative AI (GenAI):** Deep learning systems that can create new content — including text, images, audio, video, and code — based on patterns learned from large datasets.

- **Online Information Ecosystem:** The entire environment in which information is created, shared, and consumed on the internet.
  - (Gen) AI adds an additional layer potentially impacting each of these stages.
  - I will focus on platforms for news, search, user-generated content etc.

Conceptual Framework: GenAI and Online Information Ecosystem



1. Production of Content

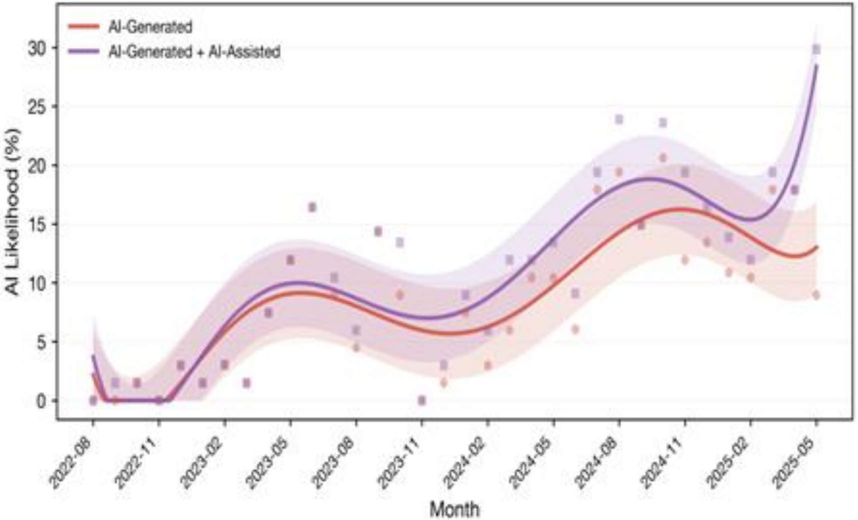
The Extent of AI-Generated Text on the Internet (Dolezal, Alam, Graham, and Bohacek 2026)

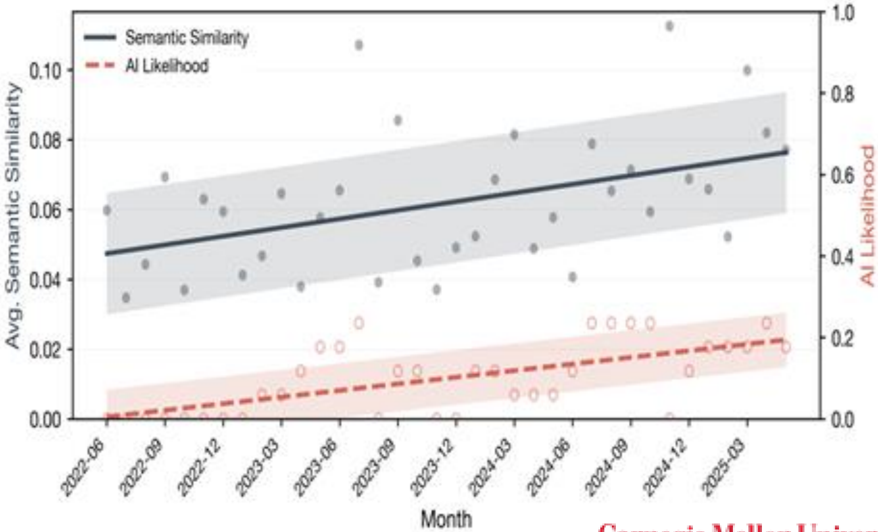
- Representative longitudinal sample of websites from the Internet Archive's Wayback Machine,
- Covering monthly intervals from August 2022 to May 2025.
- Stratified sampling across factors such as time, URL depth, and top-level domain to reduce archival bias.
- Only English-language pages with at least 100 words were retained for analysis.
- Use Pangram to differentiate between AI and non-AI text.

The Extent of AI-Generated Text on the Internet (Dolezal, Alam, Graham, and Bohacek 2026)

- By mid-2025, about 35% of newly published websites are classified as AI-generated or AI-assisted
- AI-generated text had lower semantic diversity and more positive sentiment (no evidence of decreased factual accuracy.)

Caveat: this is a new paper and limited details about the sample.





Use of GenAI impact on News Production

- Audited 250K+ U.S. newspaper articles (2022–2025) using the Pangram AI detector.

- Found that ~9% of recent newspaper articles were AI-generated or AI-assisted.
  - common in smaller local outlets and topics like weather, tech, and health.
- Limited Transparency: Manual audit of AI-flagged articles, 96% of authors did not disclose AI use.
- AI-generated articles were significantly more error-prone: 41% contained hallucinated or false claims.

AI use in American newspapers is widespread, uneven, and rarely disclosed

Jenna Russell, Marzena Karpinska, Destiny Akinode, James Zhou
Katherine Thai, Bradley Emi, Max Spero, Mohit Iyyer
University of Maryland, College Park; Simon Fraser University; UMass Amherst; Pangram Labs

Why these exercises might matter?

- Important as a descriptive exercise at a time of technological change.
- Higher quantity with limited human attention can have downstream issues (AI slop)
- Undercurrents of quality issues – but similar concerns about UGC dominating the internet.
- Synthetic data will be part of training data for LLMs down the road.

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Ideology and Composition Among an Online Crowd: Evidence from Wikipedians

Shane Greenstein, Grace Gu, Feng Zhu

Published Online: 24 Sep 2020 | https://doi.org/10.1287/mnsc.2020.3661

Home > Management Science > Vol. 70, No. 9 >

(How) Does User-Generated Content Impact Content Generated by Professionals? Evidence from Local News

Ananya Sen, Tom Grad, Pedro Ferreira, Jörg Claussen

Hyper-personalization possible

- Cost of production decreases – can potentially be used to hyper-personalize.
- In an audit study of email campaigns in the 2024 US election cycle, we find almost no personalization (Chu et al., 2025)
- Chopra et al. (2025) develop an app to customize news at scale for their study.
- But what are the returns to hyper-personalization? Is there demand for it?

Data-Driven Auditing of Black-Box AI Systems: An Application to Political Campaigns

31 Pages • Posted: 16 Dec 2024 • Last revised: 14 Jun 2025

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Date Written: December 15, 2024

News Customization with AI

Felix Chopra Ingar Haaland Fabian Roeben
Christopher Roth Vanessa Sticher

November 8, 2025

Flipside: Misinformation and Manipulation Concerns

- Reduced cost of producing misinformation – “flood the zone”
- Definite increase in the quality of synthetic content (Campante et al 2025).
- Can be used as an epistemic shield (work in progress)
- But, LLMs can also be used in scaling up fact-checking.

Charting the Landscape of Nefarious Uses of Generative Artificial Intelligence for Online Election Interference

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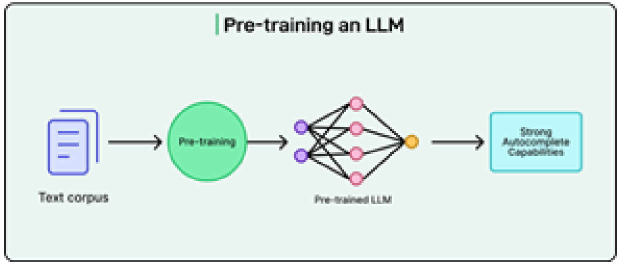
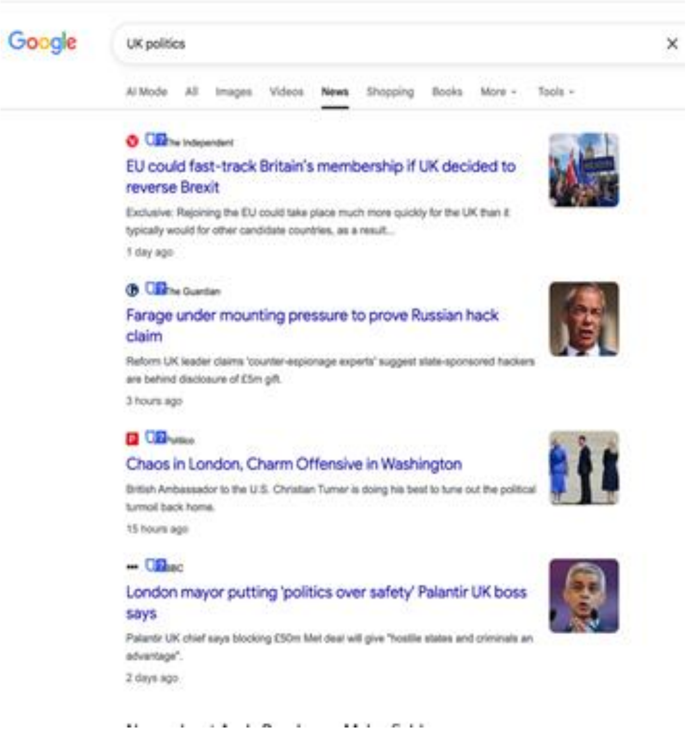
COMMENTARY

Misinformation reloaded? Fears about the impact of generative AI on misinformation are overblown

2. Distribution Stage



Google News



- **Symbiotic Relationship:** Search engines need content; publishers need visibility.
- There is a literature on disputes between news publishers and search platforms.
- Using shutdowns (Spain) and opt-outs (Germany) from Google News (Calzada and Gil, 2019) show that Google News snippets and publisher (news) traffic are complements.

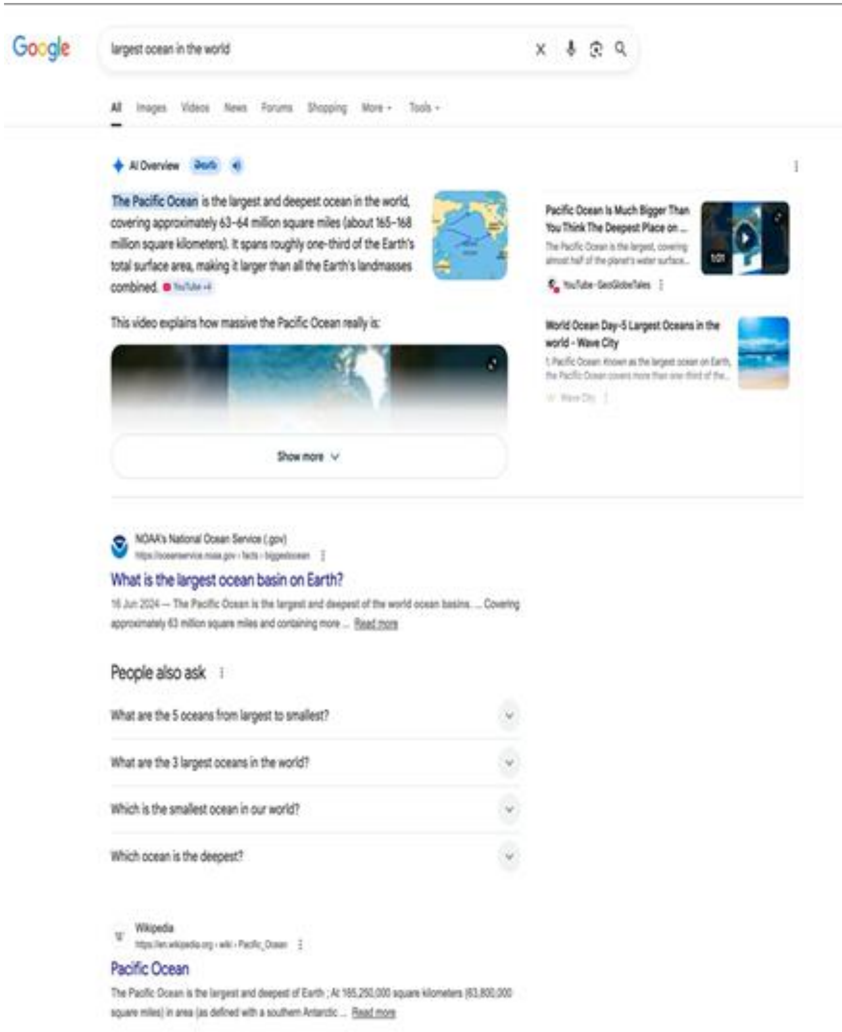
Google AI Overviews and Publisher Traffic: Evidence from a Field Experiment

Saharsh Agarwal
ISB

Ananya Sen
CMU

Google AI Overviews and Publishers

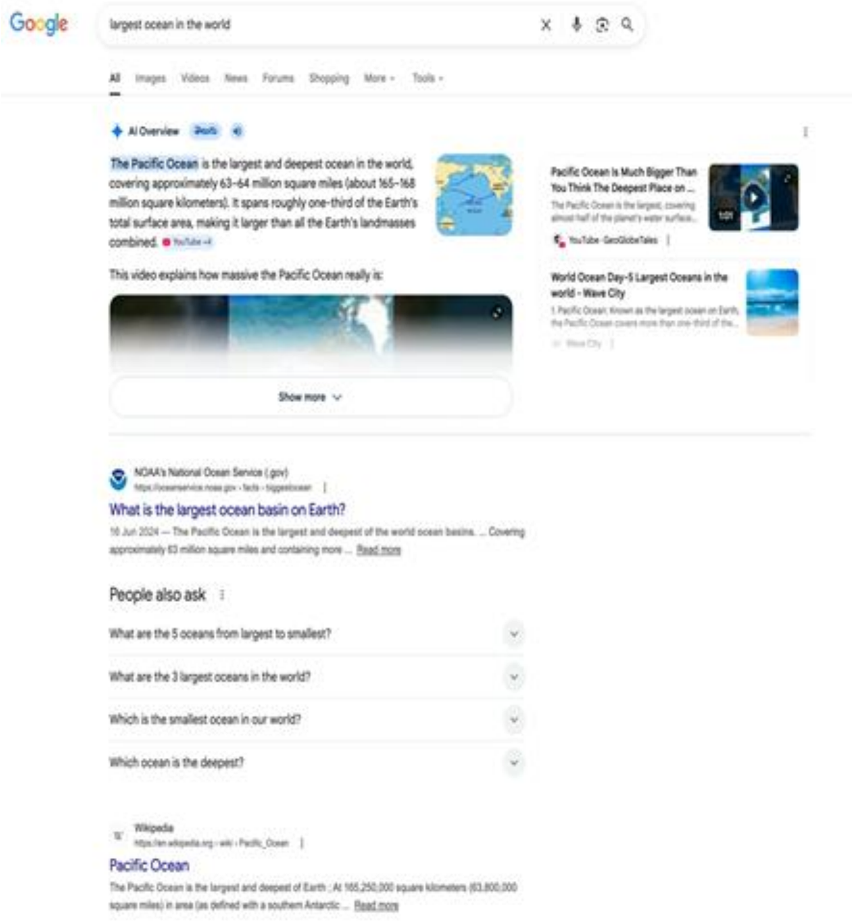
- **The Shift:** Google “AI Overviews” (AIOs) provide GenAI summaries directly on the search engine results page (SERP).
- **The Concern:**
  - Do AIOs “steal” traffic by satisfying user intent on-page?
  - Open questions related to click quality.
- **Policy Stakes:** UK CMA, US Sherman Act, and publisher lawsuits.



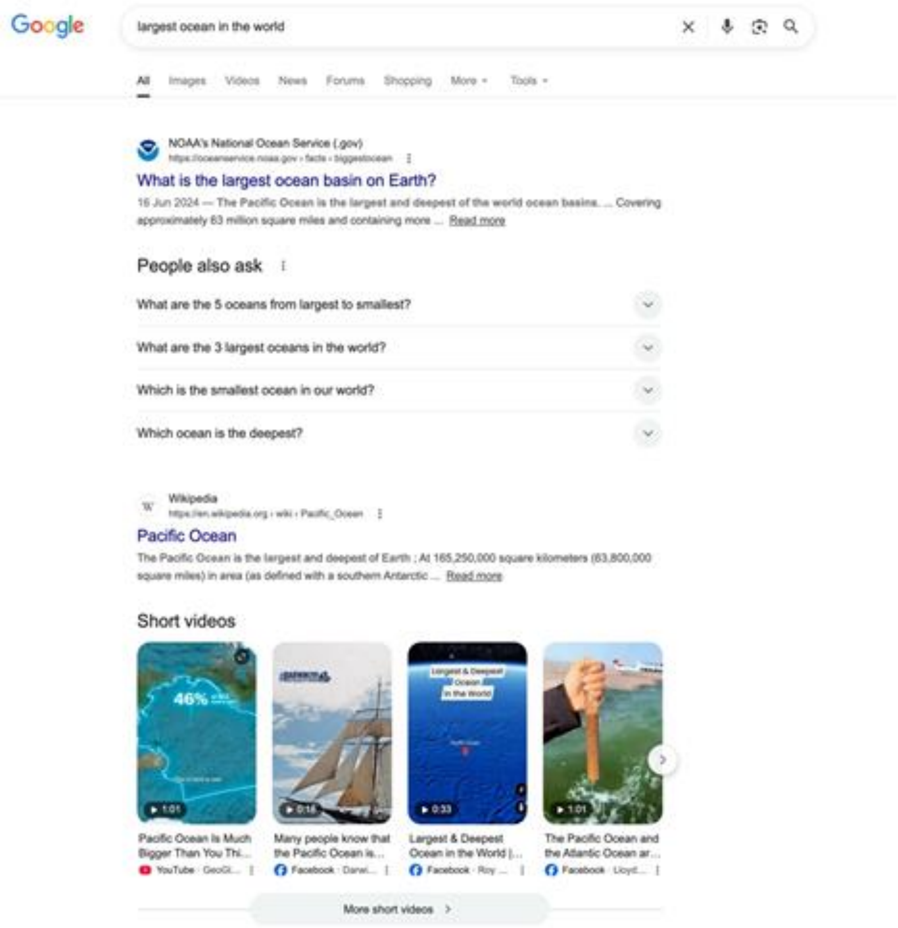
This paper

- **Method:** Randomized Field Experiment ($N = 1, 065$).
- **Tool:** Custom Chrome Extension to modify Google SERP in real-time.
- **Treatments:**
  1. **Control:** Status quo (AIOs shown).
  2. **Hide AIO (HAIO):** AIOs removed, organic results shifted up seamlessly.
- **Measurement:** Passive tracking of URL visits, clicks, scrolls, and an endline survey.

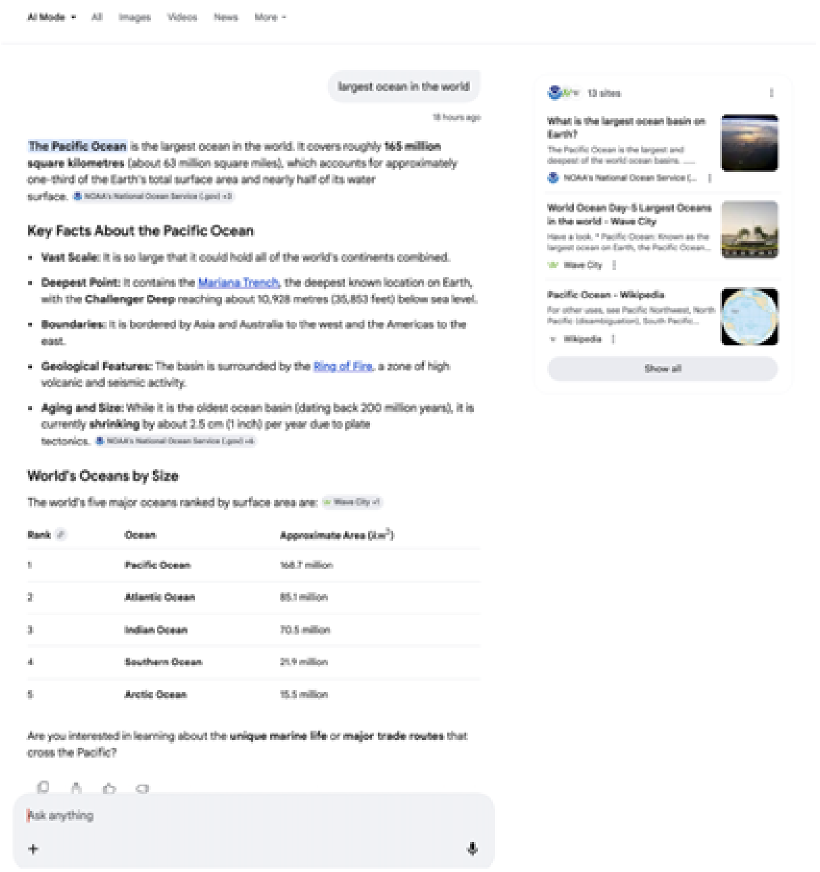
Control: Show AIO (HAIO)



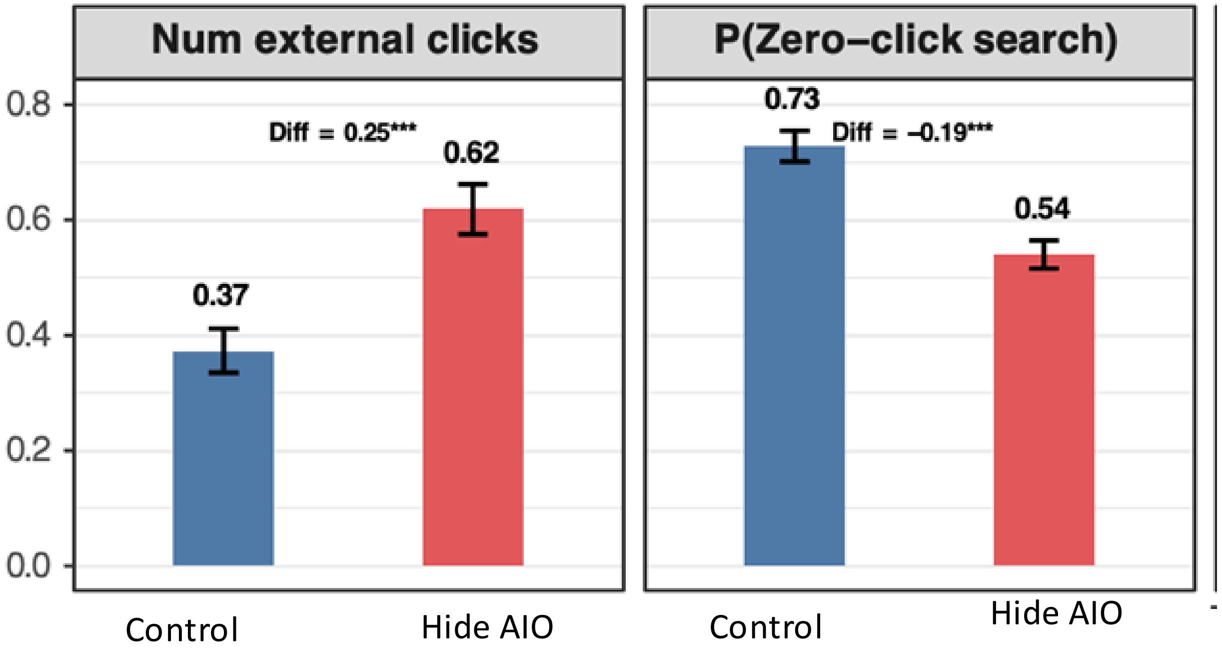
Treatment 1: Hide AIO (HAIO)



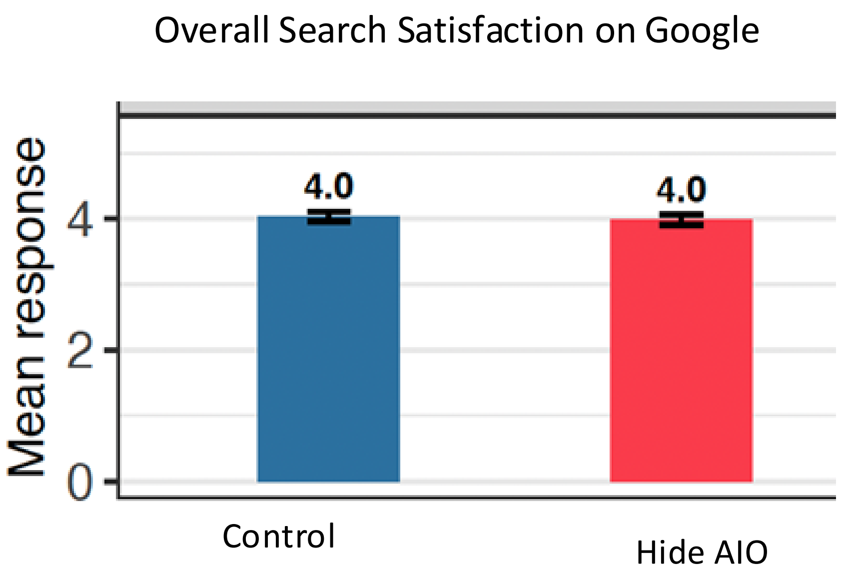
Treatment 2: AI Mode

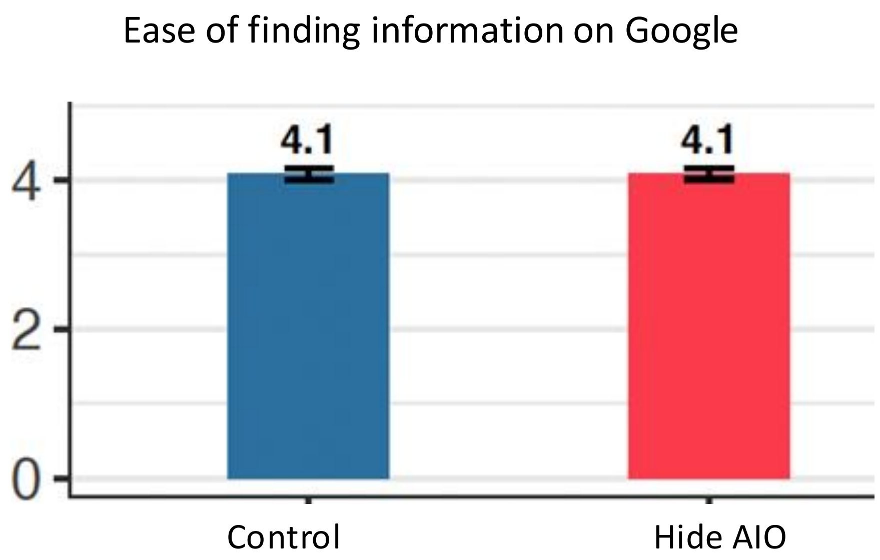


Takeaway 1: Organic Clicks ⬆ Zero Clicks Probability ⬇



Takeaway 2: User Satisfaction Unaffected (Endline Survey)





- AI summaries reduce downstream publisher traffic without improving user satisfaction.

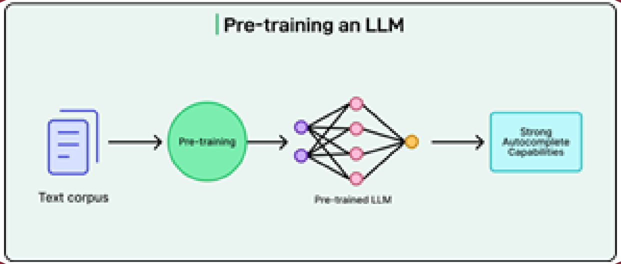
Takeaway 3: Downstream Time Spent Unaffected (Conditional on Click Analysis)

VARIABLES	(1) Bounce	(2) Bounce w/o engage	(3) Log(duration)
HAIO	-0.00218	-0.00193	-0.0438
	(0.0116)	(0.0114)	(0.0484)
From within AIO	-0.00632	0.00136	-0.00901
	(0.0231)	(0.0231)	(0.0872)
Constant	0.215***	0.207***	3.606***
	(0.00847)	(0.00832)	(0.0343)
Observations	37,349	37,349	37,338
R-squared	0.006	0.005	0.012

(1) Treatment vs. All Clicks in Control

(2) Treatment vs. Clicks coming from AIO in Control

2. Distribution Stage



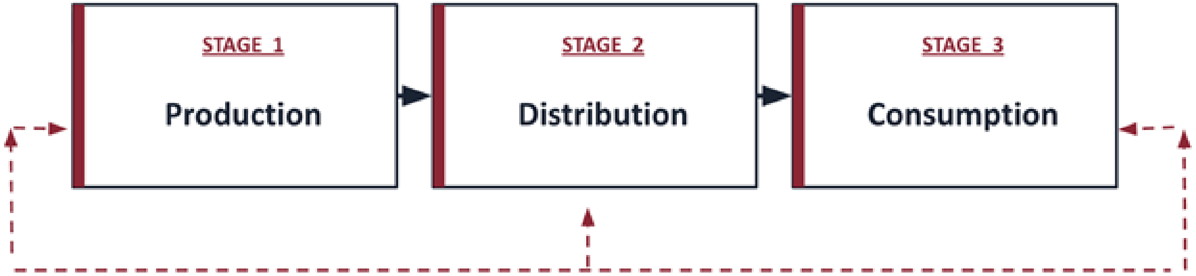
Publisher's Adaptation to the New Distribution

Adverse Selection in the AI Data Commons

60 Pages • Posted: 9 Apr 2026

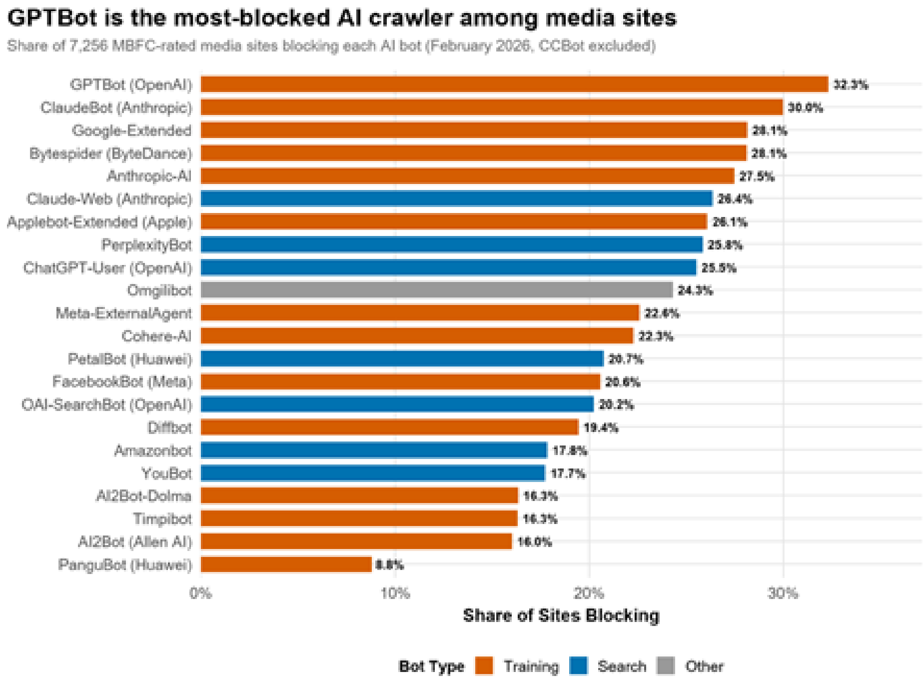
Kai Zhu
Bocconi University

- **Big Picture Recap**
  - Generative AI systems depend heavily on high-quality web content for training.
  - Content producers (news outlets, publishers etc) are usually not compensated for AI training usage.
- **Impact of Blocking on Feedback Loop**
  - High-quality publishers have incentive to block AI crawlers due to potential copyright violations.



Publisher’s Can Strategically Withdraw

- Zhu (2026) analyzes 9,611 media and news websites
- Examines robots.txt and crawler blocking behavior




```
User-agent: CCBot
Disallow: /

User-agent: GPTBot
Disallow: /

User-agent: ia_archiver
Disallow: /

User-Agent: omgili
Disallow: /

User-Agent: omgilibot
Disallow: /

User-agent: Twitterbot
Allow: /*?*smid=
```

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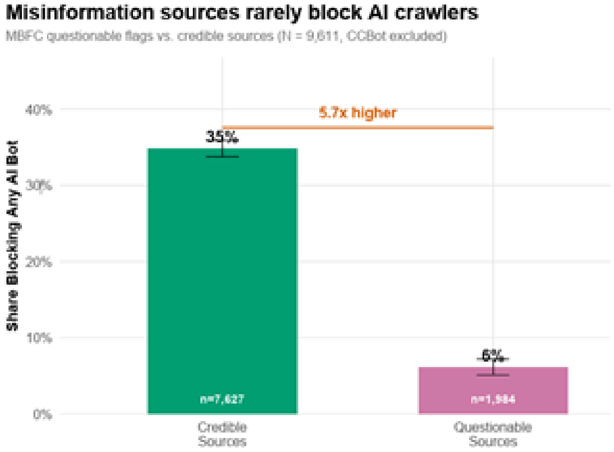
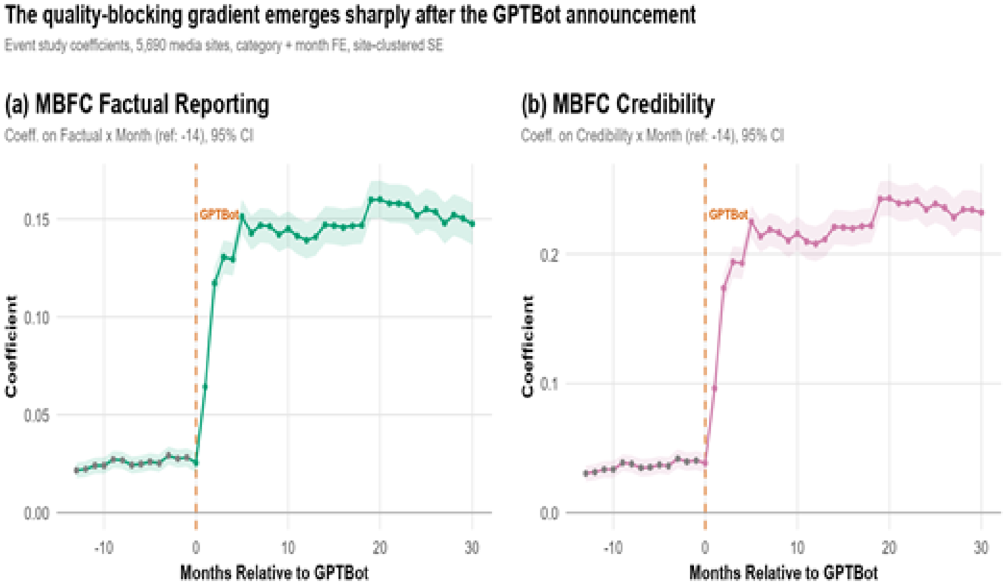
Who reacts and when?



(a) Credible vs. questionable sources



Strategic Response by State Apparatus

- LLMs exhibit a stronger pro-government stance in the languages of countries with lower media freedom than in those with higher media freedom.
- Case study on China:
  - Media scripted and curated by the Chinese state appears in LLM training datasets.
  - Prompting LLMs in Chinese generates more positive responses about China's institutions and leaders than the same queries in English.
- Training data governance becomes geopolitics.

nature

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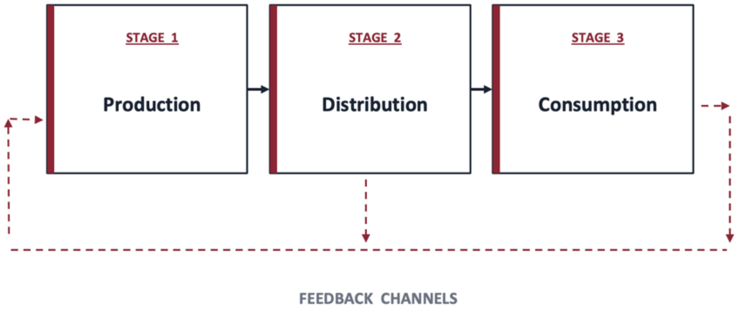
Article | Published: 13 May 2026

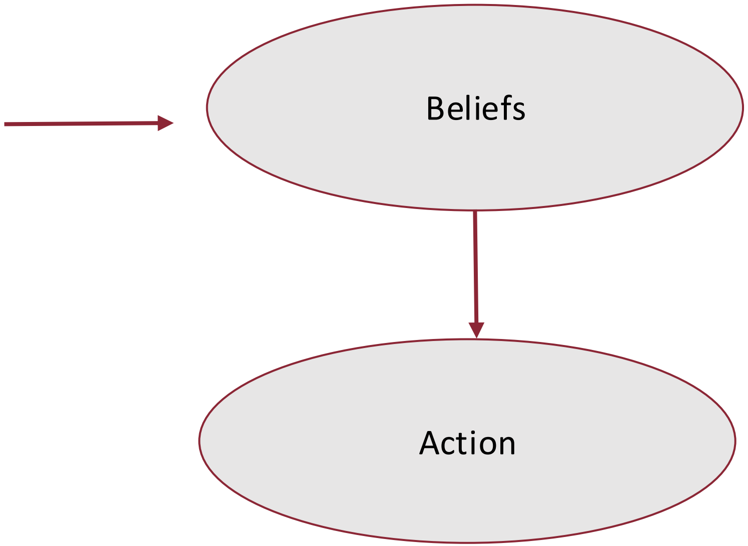
State media control influences large language models

Hannah Waight, Eddie Yang, Yin Yuan, Solomon Messing, Margaret E. Roberts, Brandon M. Stewart & Joshua A. Tucker

Nature (2026) | Cite this article



3. Consumption



Conversations with LLMs and its impact on beliefs

- Study personalized AI dialogues with people who hold conspiracy beliefs.
- 2000+ US based individuals engaged in three rounds of conversation with GPT-4.
- Drew from a pool of variety of conspiracy theories but each individual engaged with only one.
- Control group had conversations about a neutral topic.

RESEARCH ARTICLE | ARTIFICIAL INTELLIGENCE

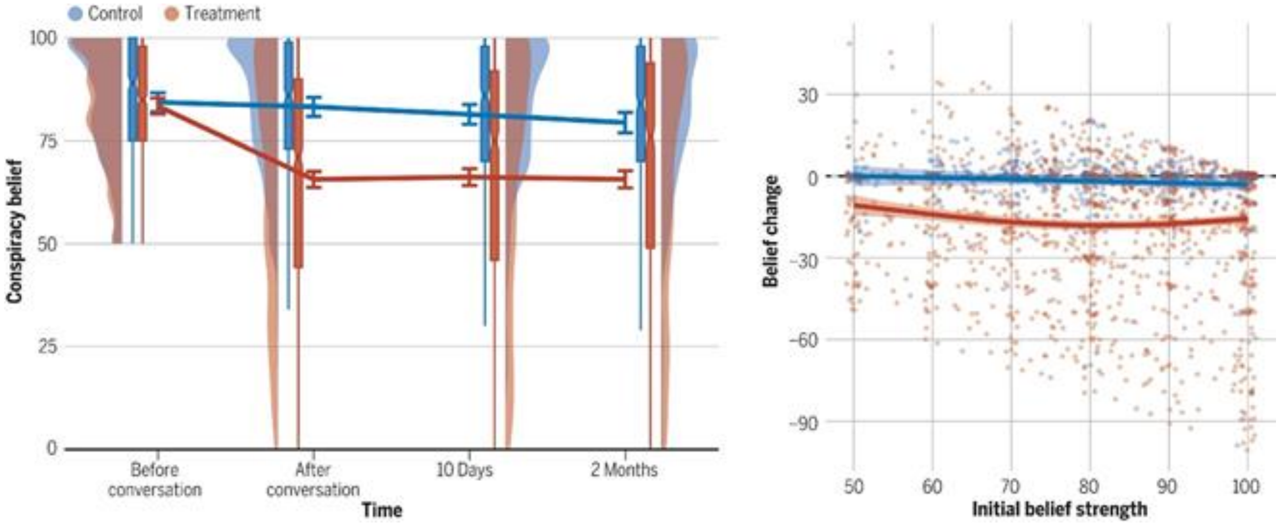
Durably reducing conspiracy beliefs through dialogues with AI

THOMAS H. COSTELLO, GORDON PENNYCOOK, AND DAVID G. RAND Authors Info & Affiliations

SCIENCE • 13 Sep 2024 • Vol 385, Issue 6714 • DOI: 10.1126/science.adq1814

Conversations with LLMs and its impact on beliefs (Costello et al. 2024)

- AI dialogue reduces conspiracy belief by 20% on average, persisting for at least two months.
- The debunking effect held even when the AI was labeled as a human expert (Boissin et al., 2025)



Treatment: AI attempted to refute the conspiracy theory; Control: AI discussed an irrelevant topic

Can AI Persuasion Translate to Action? Mixed Evidence

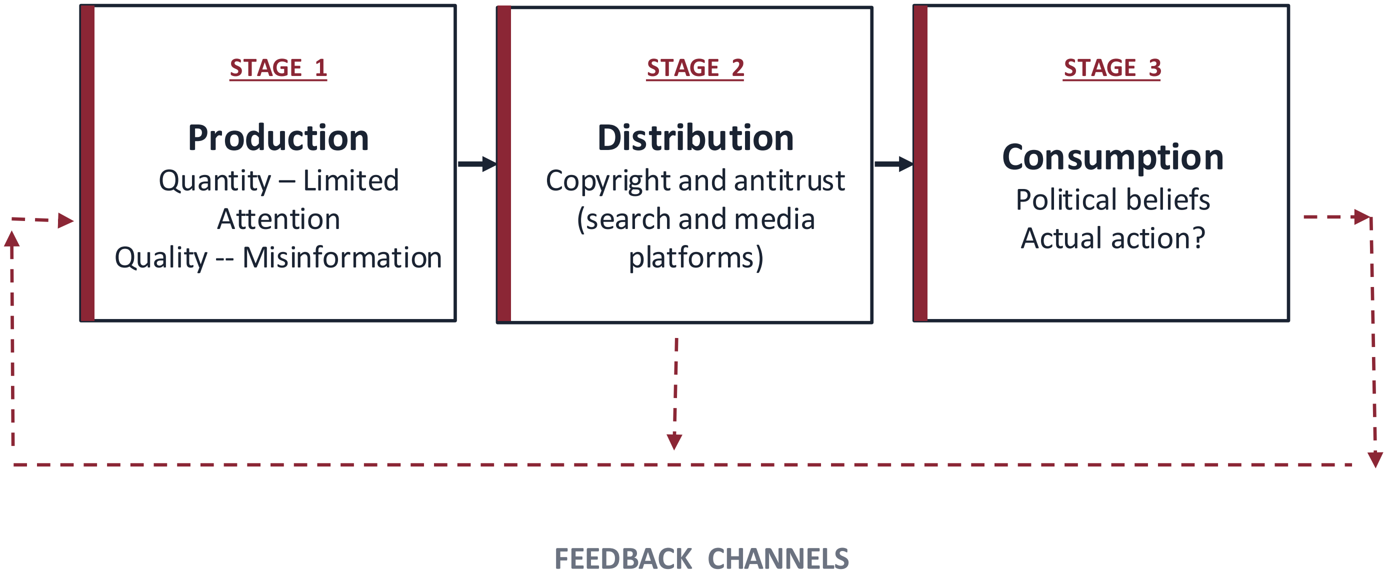
- **Within Experiment Behavior**
  - AI persuasion can translate into real behavior in experiments: conversational AI increased real petition signing and charitable donations through information provision (Hackenburg et al., 2026)

- **Outside Actions**
  - AI chatbot lowered the cost of information acquisition but no clear effect on self-reported turnout or the direction of voting (Ash, Galletta & Opocher, 2025)

Some thoughts

- What is the relevant counterfactual here?
- Access to information prior to GenAI was still relatively easy.
  - Supply-side didn't seem to be the bottleneck.
- These studies are conditional on using LLMs for conversations.
- Great start – but how do we get people to use these in the first place and solve the demand side bottleneck.

Summary


STAGE 1
Production
Quantity – Limited Attention
Quality -- Misinformation
STAGE 2
Distribution
Copyright and antitrust (search and media platforms)
STAGE 3
Consumption
Political beliefs
Actual action?
FEEDBACK CHANNELS

On Wednesday:

- A key challenge is preserving incentives for high-quality information in an AI-mediated ecosystem —

GenAI Misinformation, Trust, and News Consumption: Evidence from a Field Experiment

Appendix

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